

Project Guideline

Course: BUS 321 - Operations Management

Integration: Start for Future – Learn Program

Theme: *Sustainability for Future*

Format: Team-based (2-6) project with **pitch presentations**

1. Project Objective

Students apply **operations management concepts** to design and propose an **innovative, sustainable solution** addressing real-world sustainability challenges. The project bridges academic learning with entrepreneurial practice by following the *Learn–Build–Scale* pathway of Start for Future.

2. Learning Outcomes

By the end of the project, students will be able to:

- Apply **operations management tools** (process design, capacity planning, quality management, supply chain, lean methods) to sustainability-driven challenges.
- Demonstrate **systemic thinking** in balancing efficiency, effectiveness, and sustainability.
- Collaborate in **multidisciplinary teams** to develop entrepreneurial solutions.
- Pitch ideas effectively to diverse stakeholders.

3. Challenge Framework

Students choose a challenge within the **Sustainability for Future** domains:

- **Sustainable Production & Operations** (energy efficiency, lean & green processes, waste minimization)
- **Sustainable Services** (mobility, food systems, education, healthcare)
- **Sustainable Supply Chains** (circular economy, green logistics, fair sourcing)
- **Digital Transformation for Sustainability** (blockchain, AI, IoT for traceability and efficiency)

4. Project Process & Milestones

Phase 1: Team Formation

- Form interdisciplinary teams (4–6 members).
- Select sustainability challenge area.

Phase 2: Problem Framing

- Apply tools: *process mapping, root cause analysis (5 Whys, Fishbone)*.
- Deliverable: *Problem Statement & Initial Process Analysis (1 page)*.

Phase 3: Concept Development

- Develop solution using *Lean Canvas* adapted for sustainability.
- Apply *operations strategy* (cost, quality, speed, flexibility, sustainability).
- Deliverable: *Lean Canvas & Draft OM Strategy (2–3 slides)*.

Phase 4: Operations Design

- Model the proposed process (flowchart, capacity analysis).
- Assess resources, risks, and sustainability KPIs (e.g., CO₂ reduction, waste, cost savings).
- Deliverable: *Operations Plan Document (3–4 pages)*.

Phase 5: Validation

- Prototype concept (low-fidelity mockup, simulation, or service storyboard).
- Validate with peers and mentors.
- Refine business & operations case.

Phase 5: Pitch Preparation

- Deliverable: *Pitch Deck (10 slides max)*.

Phase 6: Final Pitch Preparation and Project Delivery

- Deliverable: *Final Pitch Deck (10 slides max)*
- Deliverable: *Final Project*

6. Assessment Criteria

- **Relevance (20%)** – Does the solution tackle a clear sustainability challenge?
- **Application of OM Concepts (25%)** – Is there evidence of process design, capacity analysis, quality or lean methods?
- **Sustainability Impact (20%)** – Are environmental, social, and economic effects addressed?
- **Innovation & Feasibility (20%)** – Is the idea original, realistic, and scalable?
- **Pitch Quality (15%)** – Clarity, structure, persuasiveness.

Student Worksheet Package

1. Problem Statement Worksheet

(Phase 1: Team Formation & Problem Framing)

Challenge Area: _____

Team Members: _____

Problem Statement

- What sustainability problem are you addressing?
- Who is affected (stakeholders)?
- Why is it important now?

2. Initial Analysis Worksheet I

Root Cause Analysis (5 Whys / Fishbone)

- Main problem: _____
- Why? 1 → _____
- Why? 2 → _____
- Why? 3 → _____
- Why? 4 → _____
- Why? 5 → _____

2. Initial Analysis Worksheet II

Process Mapping

- Current process flow (draw boxes/arrows)
- Identify bottlenecks, inefficiencies, or unsustainable practices.

3. Sustainability Lean Canvas (adapted)

(Phase 2: Concept Development)

Block	Guiding Questions	Notes
Problem	What are the top 3 sustainability-related problems?	
Customer Segments	Who experiences these problems?	
Unique Value Proposition	Why is your solution impactful and different?	
Solution	What is your idea/approach?	
Channels	How will it reach stakeholders/users?	
Key Metrics	What sustainability KPIs will you track? (e.g., CO ₂ saved, waste reduced)	
Revenue Streams	How could this solution be financially viable?	
Cost Structure	What resources/operations costs are required?	
Sustainability Impact	Environmental, social, and economic benefits	

4. Operations Design Worksheet

(Phase 3: Operations Design)

A. Process Flowchart

- Sketch proposed process (steps, inputs, outputs).

B. Capacity Analysis

- Key resources needed (equipment, people, tech): _____
- Bottleneck capacity (units/hour or tasks/week): _____

C. Quality & Lean Methods

- How will you ensure quality and reduce waste? (e.g., 5S, Kaizen, TQM)

D. Sustainability KPIs

- Energy savings: _____
- Waste reduction: _____
- CO₂ emissions avoided: _____
- Social impact (jobs, inclusivity, fairness): _____

5. Validation & Feedback Worksheet

(Phase 4: Validation)

A. Prototype / Mockup Description

- Sketch or describe your solution prototype.

B. Stakeholder Feedback

- Who gave feedback? _____
- Key insights / suggestions: _____

C. Refinements

- Based on feedback, what will you improve?

6. Pitch Deck Template (10 slides max)

(Phase 5: Pitch Prep)

1. Title & Team Introduction
2. Problem Statement (why it matters)
3. Current Process (inefficiencies / unsustainable aspects)
4. Your Solution Concept
5. Operations Design (process flow, capacity, resources)
6. Sustainability Impact (KPIs, triple bottom line)
7. Innovation & Feasibility (why it works)
8. Stakeholder Benefits & Business Case
9. Roadmap & Next Steps
10. Closing (call to action, team photo/logo)

7. Final Pitch & Project Delivery

(Phase 6: Final Pitch & Project Delivery)